**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 273 · Compilers, interpreters and assemblers

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What does LMC LDA do?

Q2. What is an executable?

Q3. What is a mnemonic?

Q4. Sorted [2,5,8,11,14,17,20,23,26], target23. Which midpoint indexes?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

compiler — Translates high-level source into target code.

interpreter — Runs a program through an execution runtime.

assembler — Translates assembly to machine encoding.

object code — Translator-generated machine-level representation.

## New learning

Understand — A compiler translates source code into another representation, often creating executable machine code via linking.

Apply — An interpreter executes program statements with a runtime, often without a separate standalone compiled executable.

Evaluate — An assembler translates assembly mnemonics and operands into processor machine instructions; translation strategies can overlap in modern environments.

## Worked example

A developer wants to distribute a compiled native executable. Which translator and what does it produce?

1. Predict first — A compiler translates the original language into target/object code

2. Model answer — linking resolves required components.

# We Do · Practise with guidance

A developer writes LMC-style mnemonics and wants binary instruction codes. Which translator is relevant?

```
LMC mnemonics → assembler → machine-code instructions
```

Predict first

An assembler converts symbolic instruction representations into machine instruction encodings.

Model answer

Explain the reasoning and justify the output.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Identify compiler, interpreter and assembler by typical role. [3]

### Question 2

Explain two differences between compiling native code and interpreting source at runtime. [6]

### Question 3

Evaluate using an interpreter in a classroom programming exercise versus distributing a native compiled application. [6]

# Exit Ticket · Answer before checking

An assembler translates assembly-language mnemonics into machine instructions.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Loads value into ACC from mailbox.

Q2. Machine-executable program representation.

Q3. Human-readable instruction abbreviation.

Q4. mid4→14, mid6→20, mid7→23; indexes 4,6,7.

### You Do pages · Independent questions

Q1. Compiler translates source to target code [1]; interpreter runs code using runtime [1]; assembler maps assembly to machine instructions [1].

Q2. Compiled output may be distributed separately [2]; interpreter/runtime required for interpreted execution [2]; translation timing and error reporting can differ [2].

Q3. Interactive execution/feedback [2]; convenient debugging [1]; runtime dependency [1]; compiled deliverable useful for native deployment [1]; performance/cross-platform trade-offs [1].

### Exit Ticket · Model answer

True.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding